Venkata Swamy Kalyan Nakka

3rd year PhD Student, SPIES Research Lab
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Research Interests

Adversarial Machine Learning, AI Security, and Language Models (Large and Small).

Education

Texas A&M University, College Station, TX, United States. **Jan 2024 – May 2027**
Doctor of Philosophy (Ph.D.) in Computer Science, GPA: 4.0/4.0

Texas A&M University–Kingsville, Kingsville, TX, United States. **Aug 2021 – May 2023**
Master of Science (M.S.) in Computer Science, GPA: 4.0/4.0 [Distinction]

Indian Institute of Technology (IIT) – Dhanbad, India. **Jul 2012 – Apr 2016**
Bachelor of Technology (B.Tech.) in Mechanical Engineering, GPA: 7.73/10.0 [First Class]

Academic & Professional Experience

Texas A&M University, College Station, TX, United States. **Jan 2024 – Present**
Graduate Research Assistant

Texas A&M University–Kingsville, Kingsville, TX, United States. **Aug 2022 – May 2023**
Graduate Research Assistant

Texas A&M University–Kingsville, Kingsville, TX, United States. **Jan 2022 – Jul 2022**
Graduate Teaching Assistant

Soroco Limited, Bangalore, India. **Sep 2019 – Jul 2021**
Senior Software Engineer

Infosys Limited, Bangalore, India. **Nov 2018 – Aug 2019**
Senior Software Engineer

Infosys Limited, Bangalore, India. **Nov 2016 – Oct 2018**
Software Engineer

Honors & Achievements

Distinguished Student Award (2023) – Awarded to top graduate student (only 1) across the university

Dean's Merit Scholarship (2022) – Recognized among top 2% of engineering graduate students for academic excellence

Computer Science Graduate Scholarship (2021) – Merit based scholarship for top 5% of CS graduate students

Rockwell International Scholarship (2021) – Awarded to academically top 2% of international graduate students

Insta Award (2018) – Corporate recognition for outstanding project delivery and performance excellence at Infosys

IIT MCM Scholarship (2013-2016) – Merit-based scholarship for top 20% of undergraduate students across the university

All India Rank 10760 in IIT-JEE (2012) – Nationwide top 2 % in entrance exam for IISc & IITs

All India Rank 8076 in AIEEE (2012) – Nationwide top 1% in entrance exam for NITs

Publications

Peer-Reviewed Conference/Workshop Papers

[C7] [**EACL 2026**] **Kalyan Nakka** and Nitesh Saxena. "*BitBypass: A New Direction in Jailbreaking Aligned Large Language Models with Bitstream Camouflage*", In Findings of European Chapter of the Association for Computational Linguistics (EACL).

[C6] [**IJCNLP-AACL 2025**] **Kalyan Nakka**, Jimmy Dani, Ausmit Mondal, and Nitesh Saxena. "*LiteLMGuard: Seamless and Lightweight On-Device Guardrails for Small Language Models against Quantization Vulnerabilities*", In Findings of International Joint Conference on Natural Language Processing & Asia-Pacific Chapter of the Association for Computational Linguistics (IJCNLP-AACL).

[C5] [**WiMob 2025**] **Kalyan Nakka** and Habib M. Ammari. "*Stochastic Connected k-Coverage in Planar Wireless Sensor Networks Using Optimal Hexagonal Tessellation*", In International Workshop on Selected Topics in Wireless and Mobile computing (STWiMob) of International Conference on Wireless and Mobile Computing, Networking and Communications (WiMob).

[C4] [**PST 2025**] Jimmy Dani, **Kalyan Nakka**, and Nitesh Saxena. "*A Machine Learning-Based Framework for Assessing Cryptographic Indistinguishability of Lightweight Block Ciphers*", In Annual International Conference on Privacy, Security & Trust (PST).

[C3] [**ECCE 2024**] BoHyun Ahn, **Kalyan Nakka**, Nathanial Handke, Trevor Reyna, and Taesic Kim. "*Field demonstration of Blockchain-based security for a Solar Farm*", In IEEE Energy Conversion Conference and Expo (ECCE).

[C2] [**ISGT 2024**] **Kalyan Nakka**, Seerin Ahmad, Logan Atkinson, Taesic Kim, and Habib M. Ammari. "*Post-Quantum Cryptography (PQC)-Grade IEEE 2030.5 for Quantum Secure Distributed Energy Resources Networks*", In IEEE Power & Energy Society Innovative Smart Grid Technologies Conference (ISGT).

[C1] [**ISCC 2023**] **Kalyan Nakka** and Habib M. Ammari. "*Square Tessellation for Stochastic Connected k-Coverage in Planar Wireless Sensor Networks*", In IEEE Symposium on Computers and Communications (ISCC).

Peer-Reviewed Journal Articles

[J6] [**CRYPT**] Jimmy Dani, **Kalyan Nakka**, and Nitesh Saxena. "*MIND-Crypt: A Machine Learning Framework for Assessing the Indistinguishability of Lightweight Block Ciphers Across Multiple Modes of Operation*", In Cryptography (CRYPT).

[J5] [**IEEE TIA**] BoHyun Ahn, Taesic Kim, **Kalyan Nakka** Jinchun Choi, Seerin Ahmad, Hee-Yong Kwon, Nathanial Handke, and Trevor Reyna. "*Blockchain-Assisted Cybersecurity for Solar Farm*", In IEEE Transactions on Industry Applications (TIA).

[J4] [**ACM TOSN**] **Kalyan Nakka** and Habib M. Ammari. "*Hierarchical Deployment and Square Tessellation for Connected k-Coverage in Heterogeneous Planar Wireless Sensor Networks*", In ACM Transactions on Sensor Networks (TOSN).

[J3] [**IEEE ACCESS**] Seerin Ahmad, **Kalyan Nakka**, BoHyun Ahn, Taesic Kim, Dongjun Han, and Dongjun Won. "*Blockchain-assisted Resilient Control for Distributed Energy Resource Management Systems*", In IEEE ACCESS.

[J2] [**ADHN**] **Kalyan Nakka** and Habib M. Ammari. "*An Energy-Efficient Irregular Hexagonal Tessellation-based Approach for Connected k-Coverage in Planar Wireless Sensor Networks*", In Ad Hoc Networks (ADHN).

[J1] [**JPDC**] **Kalyan Nakka** and Habib M. Ammari. "*k-CSqu: Ensuring connected k-coverage using Cusp Squares of Square Tessellation*", In Journal of Parallel and Distributed Computing (JPDC).

Preprints

[P1] [**arXiv**] **Kalyan Nakka**, Jimmy Dani, and Nitesh Saxena. "*Is On-Device AI Broken and Exploitable? Assessing the Trust and Ethics in Small Language Models*".

Thesis

[T1] [**M.S.**] **Kalyan Nakka**. "*Achieving Connected k-Coverage in Wireless Sensor Networks Using Computational Geometry-Based Approaches*", Texas A&M University-Kingsville, 2023.

Research Experience

SPIES Research Lab, Texas A&M University **Jan 2024 – Present**

- ***Risks and Vulnerabilities in On-Device Small Language Models*:** In this study, we exploited well-established trust and ethics assessments for understanding the risks and vulnerabilities in on-device Small Language Models (SLMs) deployed on smartphones. The results illustrated the significant high risks of stereotypical bias, unfairness, privacy-breaching behavior and harmful response generation of on-device SLMs. Further, we demonstrated the vulnerabilities of these on-device SLMs using vanilla prompts depicting various harmful scenarios.
- ***Safeguarding On-Device Small Language Models:*** In this work, we developed a practical on-device deployable lightweight deep learning (DL)-based guardrails for safeguarding Small Language Models (SLMs) against quantization-induced risks and vulnerabilities, by characterizing a novel threat model called *Open Knowledge Attack*. The results illustrated that our prompt guard secures on-device SLMs effectively and efficiently in terms of safety and latency assessments. Further, we practically demonstrated the mitigation of these vulnerabilities using our guardrails.
- ***Jailbreaking Aligned Large Language Models:*** In this study, we demonstrated the effectiveness of novel vulnerability, called Bitstream Camouflage, in jailbreaking aligned Large Language Models (LLMs). The results illustrate the high effectiveness of this vulnerability in jailbreaking aligned LLMs in terms of adversarial performance, generating phishing content, and bypassing guard models.

CPPES Lab, Texas A&M University–Kingsville **Aug 2022 – May 2023**

- ***Blockchain-based Cybersecurity for Photovoltaic Systems:*** We designed a Blockchain-based Cybersecurity platform for securing Photovoltaic systems against control-command and firmware-update attacks from adversaries, and developed a testbed for demonstrating defense against various real-time attack scenarios.
- ***Post Quantum Cryptography (PQC) grade Distributed Energy Resources Networks:*** Our study designed a Post Quantum Cryptography (PQC) grade IEEE 2030.5 network architecture for Distributed Energy Resources (DERs), and developed a testbed for understanding the performance of various PQC cipher suites. Also, we demonstrated real-time monitoring and control of DERs using our proposed PQC-grade IEEE 2030.5 network.

- ***Blockchain-assisted Resilient Control for Distributed Energy Resource Management Systems:*** In this study, we designed a Blockchain-based resilient control mechanism for DER Management Systems (DERMS), such that monitoring and control of DERs will not be affected by failure of DERMS. We developed a testbed for demonstrating the effectiveness of our proposed resilient control mechanism for real-time voltage and frequency control recovery scenarios.

WiSeMAN Research Lab, Texas A&M University–Kingsville **Aug 2022 – May 2023**

- ***Development of fault-tolerant and energy-efficient 2D Wireless Sensor Networks using Square Tessellation:*** We designed a Square Tessellation-based connected *k*-coverage theory and developed centralized protocols *k*-CSqu (deterministic), St-*k*-CSqu (stochastic) and Het-*k*-CSqu (heterogeneous) for 2D Wireless Sensor Networks, that ensures fault-tolerant coverage and energy-efficient network operation.
- ***Development of fault-tolerant and energy-efficient 2D Wireless Sensor Networks using Irregular Hexagonal Tessellation:*** In this study, we designed an Irregular Hexagonal Tessellation-based connected *k*-coverage theory and developed centralized protocols *k*-InDi (deterministic) and St-*k*-InDi (stochastic) for 2D Wireless Sensor Networks, that ensures fault-tolerant coverage and energy-efficient network operation.
- ***Development of fault-tolerant and energy-efficient 3D Wireless Sensor Networks using Cubic Honeycomb:*** This work designs a Cubic Honeycomb-based connected *k*-coverage theory and develops centralized protocol 3D-*k*-CuHon for 3D Wireless Sensor Networks, that ensures fault-tolerant coverage and energy-efficient network operation.

Teaching Experience

Course Designer, Texas A&M University
- CSCE 765: Network Security — Fall 2025, Spring 2026

Guest Lecture, Texas A&M University–Kingsville
- CSEN 5303: Industrial Control Systems Security — Spring 2023

Graduate Teaching Assistant, Texas A&M University–Kingsville
- CSEN 5303: Massive Parallel Computing — Summer 2022
- CSEN 5303: Foundations of Computer Science — Spring 2022

Presentations & Posters

- [**EACL 2026**] "*BitBypass: A New Direction in Jailbreaking Aligned Large Language Models with Bitstream Camouflage*", 19th Conference of the European Chapter of the Association for Computational Linguistics (EACL), Rabat, Morocco, March 2026
- [**TICI 2026**] "*BitBypass: A New Direction in Jailbreaking Aligned Large Language Models with Bitstream Camouflage*", Texas A&M Initiative for Connected Intelligence (TICI), College Station, TX, March 2026
- [**TICI 2026**] "*LiteLMGuard: Seamless and Lightweight On-Device Guardrails for Small Language Models against Quantization Vulnerabilities*", Texas A&M Initiative for Connected Intelligence (TICI), College Station, TX, March 2026
- [**GCRI 2025**] "*Safeguarding Both Mobile AI and User: Seamless and Lightweight On-Device Guardrails for Small Language Models against Quantization Vulnerabilities*", Texas A&M Global Cyber Research Institute Annual Summit (GCRI), College Station, TX, September 2025
- [**TICI 2025**] "*Trust and Ethics Gap in On-Device Quantized Language Models: Risks and Vulnerabilities*", Texas A&M Initiative for Connected Intelligence (TICI), College Station, TX, March 2025
- [**GCRI 2024**] "*Broken and Exploitable On-Device AI: A Comparative Trust and Ethics Assessment of Small Language Models*", Texas A&M Global Cyber Research Institute Annual Summit (GCRI), College Station, TX, September 2024
- [**GSERQ 2023**] "*Can Geometry Solve Complex Computer Science Problems?*", Texas A&M Graduate Science and Engineering Research Colloquium (GSERQ) Series, Kingsville, TX, USA, February 2023
- [**SunSpec 2022**] "*Potential Quantum Computing Attacks on Distributed Energy Resources and Post-Quantum Cryptography grade IEEE 2030.5*", SunSpec Alliance Annual Meeting (Virtual), Las Vegas, NV, December 2022

Fellowships

Graduate Research Assistantship, Texas A&M University (US $12,000 p.a.) — 2024–2026

Graduate Research Assistant Scholarship, Texas A&M University–Kingsville (US $6,000 p.a.) — 2022–2023

Dean's Merit Scholarship, Texas A&M University–Kingsville (US $1,000 p.a.) — 2022–2023

Graduate Assistant Scholarship, Texas A&M University–Kingsville (US $8,500 p.a.) — 2021–2023

HEERF III Student Scholarship, Texas A&M University–Kingsville (US $1,600 p.a.) — 2021–2022

Computer Science Graduate Scholarship, Texas A&M University–Kingsville (US $1,000 p.a.) — 2021–2022

Rockwell International Scholarship, Texas A&M University–Kingsville (US $1,000 p.a.) — 2021–2022

MCM Scholarship, IIT–Dhanbad (IND ₹72,000 p.a.) — 2013–2016

Media

Coverage

- "*Researchers unveil BitBypass, a new attack that bypasses LLM safety filters*", LinkedIn, October 2025. (Link)
- "*Quantized Small Language Models (SLMs) on devices respond to harmful queries directly...*", X (Twitter), May 2025. (Link)

Inclusions

- "*LG Uplus ixi-O leak undermines on-device AI security claims*", ChosunBiz, December 2025. (Link)

Community Service

Reviewer

ACM Transactions on Privacy and Security (TOPS)	2024
IEEE Energy Conversion Conference and Exposition (ECCE)	2024
IEEE Transactions on Smart Grid (TSG)	2025
IEEE Transactions on Artificial Intelligence (TAI)	2025

Sub-Reviewer

Annual Computer Security Applications Conference (ACSAC)	2024
IEEE International Conference on Distributed Computing Systems (ICDCS)	2024
ACM Conference on Computer and Communications Security (CCS)	2025

Student Mentoring

Ausmit Mondal, B.S. Student, Texas A&M University	2024–Present
Zachary Laguna, M.S. Student, Texas A&M University	2025
Venkat Nallam, B.S. Student, Texas A&M University	2026–Present

Professional Membership

Association for Computing Machinery (ACM)	2025–2026
Association for Computational Linguistics (ACL)	2026–Present

Technical Skills

Programming Languages: [Fluent] Python, [Familiar] C#, Go, SQL, C++, Java,
Machine Learning: PyTorch, Jupyter Notebooks, scikit-learn, Huggingface, Numpy, Pandas
Technologies: Docker, Kubernetes, Git, Linux, [Familiar] AWS, GCP, Microsoft Azure

References

Dr. Nitesh Saxena, Texas A&M University	✉ nsaxena@tamu.edu
Dr. Habib M. Ammari, Texas A&M International University	✉ habib.ammari@tamiu.edu
Dr. Taesic Kim, University of Missouri	✉ tkx96@missouri.edu
Dr. Maleq Khan, Texas A&M University–Kingsville	✉ maleq.khan@tamuk.edu